Abhishek Somnath Raut

AI ENGINEER

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https://github.com/abhishekeb211 |

Passionate Artificial Intelligence engineer, eager to contribute to team success through hard work, giving attention to detail and excellent organizational skills. Clear understanding of operations and goals. Experience in Automation Research & development and AI technology. Also positively impacting overall morale and productivity. Motivated to learn, grow and excel in Deep Learning and Cloud (Azure, Aws).

EXPERIENCE:

Assistant Professor (PCCOE Coding Club)

June 2025 - Present

Data Exploration Lab, Full Stack Development Lab, Theory of Computation
Airawat Supper Computer CDAC PQC algorithm Testing, Fujitsu Quantum Challenge Finalist(Ongoing)
ISRO Cross domain RAG System, Tress species detection UAV Hyperspectral & LiDAR Data,
Skoda offline Business Report PPT Generation, Skoda Third Party Payment Automation, Assimilation Technology SCM Software, Sarva Labs MOUI ZeroTrust
Project Guide: Malicious & vulnerable Code detection AI, Multimodal LLM Inference

Data Scientist (Arya Systems Startup & Freelancer)

2024- June 2025

LLM Models, Generative & Transformers Chatbot Implementation and Research Paper Transformers & Cloud (AWS, Azure, GCP), Python Windows and Cloud apps, Algorithm Optimization, Design of Experiment Optimization. And Sensitivity Analysis

2020-2022

Data Scientist Internship (Under PhD Data Scientist)

2019-2020

Topic Modeling and NLP: RNN, LSTM and Multiple Topic Map reduce Algorithms, Electric Vehicles Motor analysis and prediction of specifications from ordinary vehicles
Graphological Traits and Handwritten Text Recognition 2019
Anomaly Detection using Opencv3 and TensorFlow with pedestrian dataset 2020

EDUCATION:

MTech AIDS (Artificial Intelligence & Data Science) 8.7 CGPA — **2022-2024**
Pimpri Chinchwad College of Engineering – Pune
GENAI, LLM, & Data Science

Nanodegree Field of Study Deep Reinforcement Learning UDACITY — **2021-01**
Activities and Societies: Monte Carlo Methods, Deep Q-learning, Double Q-learning, Dueling DQN, Policy-Based Methods, Continuous Control (Actor - Critic Method) and Multi- Agent Reinforcement Learning (DDPG, MADDPG)

Nano Degree: Deep Learning & Machine Learning UDACITY — **2020-07**
Received AWS Machine Leaning Training Knowledge about GAN, Transfer Learning (TensorFlow), RNN, CNN using Pytorch & Cloud Computing

B.E. IT (Information Technology) 7 SGPA — **2016-2020**
PES MODERN COLLEGE OF ENGINEERING - Shivajinagar Pune
Machine Learning & Deep Learning

CERTIFICATIONS:

GenAI Pinnacle Program by Analytics Vidhya Issued on Apr 2024
AWS: AI Foundation Course by Udacity Issued on Oct 2021
Microsoft Certified: Azure AI Fundamentals Issued Aug on 2021
AWS: Reinforcement Learning DeepRacer Issued on 2020

On-Going Projects:

1. **Secure Cross domain-adaptive RAG + PPT Generation**
   Implemented high-speed multi-stage retrieval (domain routing + n-gram expansion, hybrid BM25+dense ANN, optional knowledge graph/SEO ranking and advanced reranking) with strict version governance, RBAC, encryption, and end-to-end audit traceability down to page/paragraph.
2. **Best Friend Forever (SAMANVAY-7) — Privacy-First Offline AI Mental Health Pod**
   Designed and developed an offline, zero-cloud AI therapy system using a local LLM (Llama 3.2) with multimodal voice/text interaction and avatar-based companionship. Implemented anonymous session orchestration, personality/context modeling and crisis override protocols with optional human escalation.
3. **Coding Agent CLI**
   Offline-first, air-gapped GenAI security scanning and remediation CLI that orchestrates scanners, normalizes SARIF/CWE findings, and uses a local LLM to triage issues and propose fixes under policy-as-code governance.

AWARDS / ACCOMPLISHMENTS:

[1] Fijitsu Quantum Computing Challenge 2026 Final Round (OnGoing)
[2] Top 10 Mumbai Hacks 2025 & Airwat CDAC Supper Computer
[3] Analytics Vidhya Machine Learning Prediction Challenge 3rd Winner) 90k Price Money Oct 2023
[4] AWS DeepRacer League License and Certification (**World 25th Rank & Top 10 in Asia and India**)
[5] AWS 600$ Scholarship (Machine Learning and Deep Learning)
[6] SIH 2020 Hardware Edition Winner (Aerodynamics Helicopter Prototype) and SIH 2020 Software Edition Runner Up (Women Safety and Child Trafficking using Cyber Space Application)
[7] ABU ROBOCON 2019 Top 25,ABU ROBOCON 2018 Rank & ABU ROBOCON 2017 Rank 7 (Autonomous Robot & Manual Robot)

RESEARCH PAPERS:

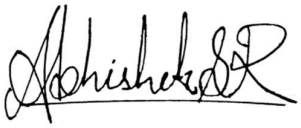
[1] The Future of Chatbots: Survey on Generative Pretrained Models, Ernie, and Gemini, ICICET 2024 doi: 10.1109/ICICET59348.2024.10616356.
[2] Prediction of mechanical properties of aluminum metal matrix hybrid composites synthesized using Stir casting process by Machine learning https://doi.org/10.1016/j.matpr.2022.04.316
[3] Research Paper: Personality analysis through handwriting recognition 2020 Journal of Discrete Mathematical Sciences and Cryptography https://doi.org/10.1080/09720529.2020.1721856

SKILLS:

[1] **Expert:** C, C++, Python, SQL and Data Analyst
[2] **Advanced:** Azure, Aws, Machine Learning & Deep Learning
[3] Statistics & Exploratory Data Analysis

LANUAGES: **English, Hindi, Marathi.**

HOBBIES: Rubik's cube, Guitar music, Instagram page Clickipng__it



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